NICOLAS SERRANO

Software Engineer
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SUMMARY

Senior Software Engineer with 7+ years of experience building systems including 4 years at the Ethereum Foundation Privacy Stewards of Ethereum (PSE) team. Core team member and maintainer of p0tion (forked 80 times), led frontend development of the KZG Ceremony, built MACI integrations with Aragon & Gitcoin.

SKILLS

Full-Stack: Typescript - Node (NestJS), Go (Gin), PostgreSQL, MongoDB, MySQL, Redis, React, PWA.
Contracts: Solidity, Hardhat, Foundry, ethers, Viem, IPFS, TheGraph, go-ethereum, ERC4337, ERC20, UUPS.
Ethereum: L1/L2s, EVM, MPTs, RLP encoding, home staker (Linux, Ethrex + Lighthouse from source).
DevOps: AWS (S3, EC2, Lambda, Amplify, etc), Docker, GitHub Actions (CI/CD), Firebase, GCP.
Agentic Engineering: Claude Code, Codex, hooks, skills, spec-kit, test-driven planning, Ralph loops.

EXPERIENCE

Forward Deployed Engineer | Innowork S.A.S.
July 2026 – Present (5 months) | Ecuador

- Designed and developed a DAO to manage public sector funds of the Prefecture of Azuay through citizen proposals. Wrote the smart contracts in **Solidity**; unit, E2E and fuzz tests using **Foundry**. Built a coordinator backend service using **Go (Gin)**, **Postgres** and the OpenZepellin (**Rust**) relayer. Built frontend with **React**, **Wagmi**, **Privy and non-custodial wallet** options to make the UX seamless.

Software & ZK Engineer | Ethereum Foundation — Privacy & Scaling Explorations (PSE)
July 2022 – July 2026 (4 years) | Remote

Trusted Setups Team | Staff Software Engineer

- Built and maintained the **p0tion toolkit**: the most widely used trusted setup framework in Ethereum (backend, frontend, CLI, unit tests and e2e tests). The platform **hosted 20+ finalized production ceremonies** and **16,000 users** (ceremony.pse.dev). I monitored the progress and was on-call for incidents. Stack used: **Typescript**, **Ethers, SnarkJS, Firebase,** cloud functions**, AWS S3** storage.
- Led frontend development for the **KZG Ceremony** (ceremony.ethereum.org), critical component to enable Danksharding (EIP-4844), **achieving 141,416 contributors** globally over 208 days. The **largest multi-party computation of its kind** at time of publication. Stack used: **Typescript, React, Ethers**, crypto library wrapper in **Rust** for efficient and securte client-side secret generation, **IPFS** and **PWA** for censorship resistance, **ENS domains**. Deployed on **AWS** (Amplify and Cloud Monitoring) and **Pinata**.

Private Transfers Team | Senior Software Engineer

- Led system design, metrics collection, frontend development, and evaluation review for a private transfers report on Ethereum: **55 protocols evaluated across architectures** (UTXOs, stealth addresses, mixers, private L2s, FHE, TEEs, etc) and **35 properties** (CROPS, compliance, etc). Stack used: **TheGraph** and **RPC nodes**, **React**, **Tailwind**, **Claude** citations: https://private-transfers.pse.dev/
- Designed and built a private transfers zkWormholes protocol (**Private Proof of Burn**) with plausible deniability and client-side proving (CSP). Stack used: Superchain OP **ERC20** framework, **Solidity, Foundry, proxy patterns, Typescript, React, Vite, Wagmi, Circom, Circomkit, SnarkJS**.
- Developed 2 post quantum private transfers protocols. One based on **FHE** + **ORAM** (confidential amounts) and another **notes-based shielded pool**. Stack used: **Typescript, NextJS, React, Solidity**, **OpenZeppelin and Hardhat**. AI tools: Ralph loops, Claude Code agents with custom skills and hooks.

Private Voting Team | Senior Software Engineer

- Core contributor to MACI: **+596 stars** private voting protocol proposed by Vitalik, used by Gitcoin, Aragon & Devcon. Built the coordinator: a **TypeScript NestJS backend service** orchestrating tallying processes with **KMS**, versioned API & Swagger docs. Used **ZeroDev** for gasless txs (ERC-4337). Contributed to **TypeScript** SDK, **Solidity** contracts (**Hardhat**), and **Circom** ZK circuits. Wrote unit tests and e2e tests. Addressed vulnerabilities found in **HashCloak audit**.
- Led integration of MACI private voting plugin for **Aragon OSx**: **Solidity/Foundry** contracts, **Typescript NextJS**, and documentation. Onboarded and **mentored a team member** through the MACI codebase.
- Led architecture research for MACI v4 decentralization: threshold encryption and DKG for coordinator.
- Coordinated the Gitcoin GG24 Privacy Round in collaboration with the external Privote team. Supported development (MACI SDK) and reviewed applications. **Round raised 35.41 WETH** in the voting contract.

Staff Smart Contract Engineer | CEDIA — National Research & Academic Consortium of Ecuador

September 2021 – August 2022 (1 year) | Cuenca, Ecuador

- Built e-voting system used in **15 university elections** (**+150k voters**). Owned on-chain component: private, decentralized & auditable. Stack: **Go (Gin), go-ethereum, Solidity, IPFS, PWA, Docker.**
- Provisioned **authenticated RPC** infra for Ethereum, Avalanche and Polygon on Linux VMs with Nginx.

Founder | Innowork S.A.S.

August 2020 – Present (on the side) | Ecuador

- Lead **team of 3-5** (developers, product managers and designers) to deliver high-quality solutions for clients pain points. Talked and negotiated with customers. Acted as project manager in multiple projects. Owned **15+ websites & e-commerce developmen**t. Stack: Wordpress, PHP, MySQL, Stripe, Paypal.

Tech Lead & Co-Founder | Innomaps — Geoanalytics Startup

December 2017 – August 2022 (4 years 8 months) | Ecuador

- Co-founded geoanalytics startup recognized as Top 35 of 500+ projects by the Hult Prize; grew to **2-5 team members**, **5+ active corporate clients**, **50+ monthly users**, **$40,000 in revenue**. Stack: **Express, PostgreSQL, PWA, Python ML pipeline, Google ML,** Google Maps API integration.

EDUCATION

Information Technology Engineering — Cum Laude | Yachay Tech University

2015 – 2021 | Urcuquí, Ecuador

Thesis: Data Marketplaces, Blockchain & HE.

Paper: A P2P Private Data Marketplace (13 citations).

Certifications:

- Zero Knowledge Proofs MOOC (Stanford & UC Berkeley - 2023)
- Blockchain: Disruptive Technology (MIT - 2022)
- Startup Accelerator Program (Hult Prize Foundation - 2019)

WRITINGS & TALKS

- Auditor viewing keys in privacy systems. Personal blog, July 2026
- Privacy in Lending Protocols. Personal blog, July 2026
- State of Private Voting 2026. PSE Blog, November 2025
- Learnings from the KZG Ceremony. PSE Blog, 2023
- Blockchain: Sustainable Development or Energy Consumption? Bionatura Journal, 2018. 7 citations.
- IC3 Blockchain Summer Camp — 6-year consecutive participant and EF speaker alongside Arbitrum, Chainlink, JP Morgan. Given talks in Cornell University, Cornell Tech and Princeton University. IC3 Hackathon 2022 first place winner team (Smart account heuristic calculator)
- Speaker at: Encryption Day (ARG, 25), ProgCrypto (IST, 23), Aleph ZK Week (ARG 24).
- ETH Ecuador — active member (15 people) team organizing courses, technical projects, and events.